

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Friday 19 May 2023

Afternoon

Paper reference **8FM0/22**

Further Mathematics

Advanced Subsidiary

Further Mathematics options

22: Further Pure Mathematics 2

(Part of option A only)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
- Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. The operation $*$ is defined on the set $G = \{0, 1, 2, 3\}$ by

$$x * y \equiv x + y - 2xy \pmod{4}$$

- (a) Complete the Cayley table below.

(2)

*	0	1	2	3
0				
1				
2				
3				

- (b) Show that G is a group under the operation $*$

(You may assume the associative law is satisfied.)

(3)

- (c) State the order of each element of G .

(2)

- (d) State whether G is a cyclic group, giving a reason for your answer.

(1)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)

2. A linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is represented by the matrix

$$\mathbf{M} = \begin{pmatrix} 5 & 1 \\ k & -3 \end{pmatrix}$$

where k is a constant.

Given that matrix $\mathbf{M}$ has a repeated eigenvalue,

(a) determine

(i) the value of k

(ii) the eigenvalue.

(6)

(b) Hence determine a Cartesian equation of the invariant line under T .

(2)



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Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 8 marks)



3. A complex number z is represented by the point P on an Argand diagram.

Given that

$$\arg\left(\frac{z - 4 - i}{z - 2 - 7i}\right) = \frac{\pi}{2}$$

- (a) sketch the locus of P as z varies, (2)
- (b) determine the exact maximum possible value of $|z|$ (5)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 7 marks)



4. A student takes out a loan for £1000 from a bank.

The bank charges 0.5% monthly interest on the amount of the loan yet to be repaid.

At the end of each month

- the interest is added to the loan
- the student then repays £50

Let U_n be the amount of money owed n months after the loan was taken out.

The amount of money owed by the student is modelled by the recurrence relation

$$U_n = 1.005U_{n-1} - A \quad U_0 = 1000 \quad n \in \mathbb{Z}^+$$

where A is a constant.

- (a) (i) State the value of the constant A .

- (ii) Explain, in the context of the problem, the value 1.005

(2)

Using the value of A found in part (a)(i),

- (b) solve the recurrence relation

$$U_n = 1.005U_{n-1} - A \quad U_0 = 1000 \quad n \in \mathbb{Z}^+$$

(5)

- (c) Hence determine, according to the model, the number of months it will take to completely repay the loan.

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

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(Total for Question 4 is 9 marks)



5. (i) Making your reasoning clear and using modulo arithmetic, show that

214^6 is divisible by 8

(3)

- (ii) The following 7-digit number has four unknown digits

$\boxed{a}5\boxed{b}8\boxed{a}\boxed{b}0$

Given that the number is divisible by 11

- (a) determine the value of the digit a .

(2)

Given that the number is also divisible by 3

- (b) determine the possible values of the digit b .

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 8 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS

